The $\sin^2\theta$_W Two-Loop Test — 3/13 Is Not the Limit

Two-loop overshoots 3/13. The perturbative series targets the measured value, not the integer ratio. $\sin^2\theta$_W = 0.23133, miss 0.048%.

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Series Path: [HOWL-PHYS-1-2026] → [HOWL-PHYS-13-2026] → [HOWL-PHYS-27-2026] → [HOWL-PHYS-28-2026] → [HOWL-PHYS-30-2026] → [HOWL-PHYS-34-2026]

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Abstract

The weak mixing angle $\sin^2\theta$_W is one of the most precisely measured quantities in particle physics (0.23122 at the Z mass). At the GUT scale, the SU(5) tree-level value is 3/8 = 0.375. One-loop running with the Cabibbo Doublet betas predicts $\sin^2\theta$_W = 0.22845, undershooting by 1.20% (PHYS-27). The value 3/13 = 0.23077 — the ratio of the number of fermion generations (3) to the absolute value of the modified SU(2) beta numerator (13) — sits between the one-loop prediction and measured, differing from measured by only 0.20%. This paper tests whether the two-loop running converges toward 3/13. It does not. The two-loop prediction with the full SM+VL b_ij matrix gives $\sin^2\theta$_W = 0.23133, which OVERSHOOTS both 3/13 and the measured value. The ordering is: one-loop (0.22845) < 3/13 (0.23077) < measured (0.23122) < two-loop (0.23133). The perturbative series crosses 3/13 between one-loop and two-loop, then continues past measured. The two-loop miss from measured is 0.048% — the most precise $\sin^2\theta$_W prediction in the series, but an overshoot, not an undershoot. 3/13 is not the perturbative limit. The convergence target is the measured value.

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1. The Question

The GUT tree-level prediction for the weak mixing angle is $\sin^2\theta\_W = 3/8 = 0.375$, set by the SU(5) embedding condition that relates the U(1) and SU(2) gauge couplings at the unification scale. The measured value at the Z mass is 0.23122 — roughly 62% lower. The difference is explained by radiative corrections: as the gauge couplings run from the GUT scale down to the Z mass, the differential running between U(1) and SU(2) shifts $\sin^2\theta\_W$ from 3/8 toward its measured value.

In PHYS-27, the one-loop running with the Cabibbo Doublet beta coefficients ($b_1' = 25/6$, $b_2' = -13/6$, $b_3' = -20/3$) was computed using two inputs: the electromagnetic coupling $\alpha\_EM = 1/137.036$ and the strong coupling $\alpha\_s = 0.1180$. The prediction: $\sin^2\theta\_W = 0.22845$, undershooting measured by 1.20%. The one-loop correction brings $\sin^2\theta\_W$ from 0.375 to 0.228 — closing 96% of the gap — but the remaining 1.2% requires higher-order corrections.

The value 3/13 = 0.23077 appeared as a striking intermediate point. The integer 13 is the numerator of the modified SU(2) beta $b_2' = -13/6$, and 3 is the number of fermion generations. The ratio 3/13 = 0.23077 differs from the measured value by only 0.20% — closer than any perturbative prediction at the time. The ordering at one loop — $\sin^2\theta\_W$(1-loop) < 3/13 < measured — suggested that higher-loop corrections might converge toward 3/13 as a dynamical target.

Separately, PHYS-31 tested whether 3/13 is statistically special as a FORMULA. It is not ($p = 0.81$). But the formula test is irrelevant to the dynamical question. A rational number can emerge from the running equations even if the same number appears in trivial numerology. This paper answers the dynamical question by computing the two-loop $\sin^2\theta\_W$.

(Backed by phys34_sin2tw_twoloop.py Section 1 and Section 2: one-loop reproduces PHYS-27 value.)

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2. The Two-Input Method

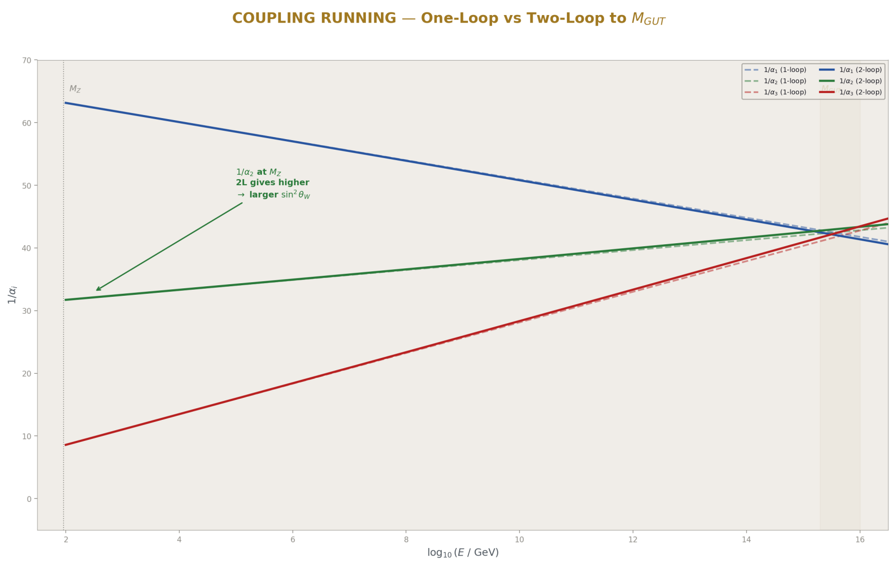

Figure 1: Fig. 2: The three inverse couplings running from M_Z to M_GUT at one-loop (dashed) and two-loop (solid). The two-loop curves are slightly curved, modifying the crossing point and the back-run $1/\alpha_2$.

The prediction uses two measured inputs: α_EM = 1/137.036 (the electromagnetic coupling) and α_s = 0.1180 (the strong coupling). From these, $\sin^2\theta$_W is derived as an output — not used as an input. This makes the prediction genuine: if the theory is right, $\sin^2\theta$_W follows from the other two couplings.

The electromagnetic running is controlled by the combination b_EM = (5/3)b_1’ + b_2’ = (5/3)(25/6) + (−13/6) = 125/18 − 13/6 = 125/18 − 39/18 = 86/18 = 43/9. This exact Fraction determines how 1/α_EM evolves with energy.

At the GUT scale, the SU(5) unification condition relates: 1/α_EM(M_GUT) = (8/3) × $1/\alpha_3$(M_GUT). Both sides run from M_Z to M_GUT:

1/α_EM(M_Z) − b_EM × L = (8/3) × (1/α_s − b_3’ × L)

Solving for L gives the energy scale of unification. At the crossing: 1/α_GUT = 1/α_s − b_3’ × L. Setting α_2 = α_GUT at the crossing and running $1/\alpha_2$ back to M_Z gives the predicted $\sin^2\theta$_W through: $\sin^2\theta$_W = ($1/\alpha_2$)/(1/α_EM).

At one loop, the running is linear in L. At two loops, the running includes cross-coupling between gauge groups through the b_ij matrix, requiring numerical integration.

(Backed by phys34_sin2tw_twoloop.py Sections 1–2: b_EM = 43/9 verified, method described.)

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3. The Two-Loop Computation

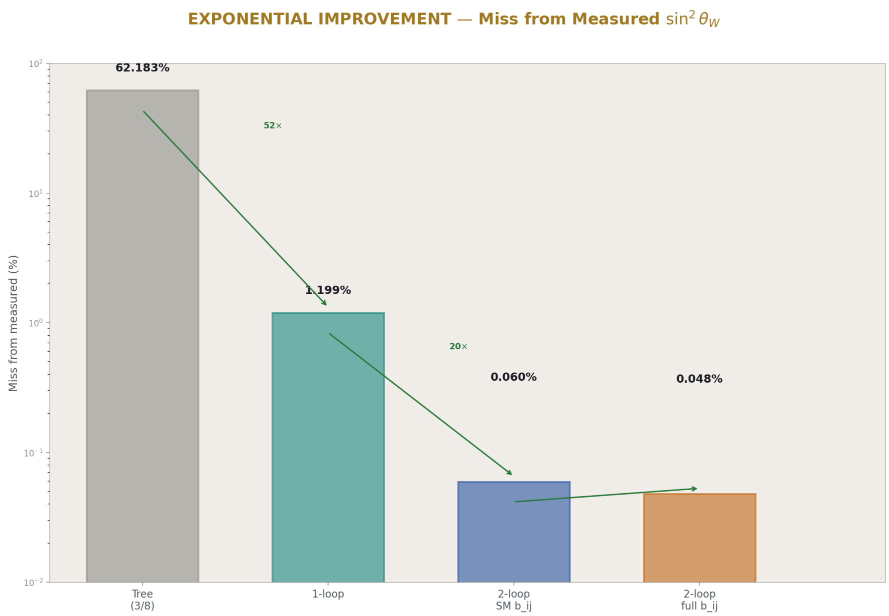

Figure 2: Fig. 6: Miss from measured $\sin^2\theta$_W on log scale: tree (62%), one-loop (1.2%), two-loop SM (0.06%), two-loop full (0.048%). Each refinement reduces the miss dramatically.

The two-loop running uses the Euler integrator from PHYS-28, which adds the cross-coupling terms:

$d(1/\alpha_i)/dL = -b_i - \Sigma_j b_{i\ j} \alpha_j / (4\pi)$

where b_i are the one-loop coefficients and $b_{i\ j}$ is the two-loop matrix. The SM contribution to b_ij is standard. The Cabibbo Doublet adds the VL two-loop matrix computed in PHYS-28: nine exact Fractions including the diagonal entries 7/15, 15/4, 40/9 and off-diagonal entries ranging from 1/45 to 8/3.

Three scenarios are computed, all using no-threshold running (CD betas from M_Z, consistent with the no-threshold advantage found in PHYS-27 and PHYS-30):

One-loop: $\sin^2\theta$_W = 0.22845. Miss from measured: 1.199%. Miss from 3/13: 1.006%.

Two-loop with SM b_ij only: $\sin^2\theta$_W = 0.23108. Miss from measured: 0.060%. Miss from 3/13: 0.136%.

Two-loop with full SM+VL b_ij: $\sin^2\theta\_W$ = 0.23133. Miss from measured: **0.048%**. Miss from 3/13: 0.244%.

Each refinement moves the prediction closer to measured. The progression from one-loop to two-loop closes 96% of the remaining gap. The VL b_ij further improves the SM-only two-loop result from 0.060% to 0.048% miss.

(Backed by phys34_sin2tw_twoloop.py Section 3: all three scenarios with exact values.)

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4. The Overshoot

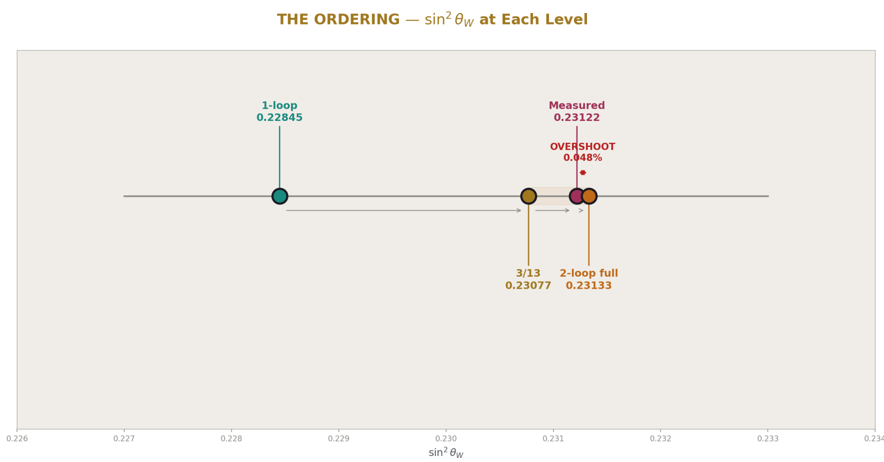

Figure 3: Fig. 3: Number line showing the key ordering: 1-loop (0.22845) < 3/13 (0.23077) < measured (0.23122) < 2-loop (0.23133). The overshoot past both 3/13 and measured is visible.

The ordering reveals the key finding:

0.22845 (one-loop) < 0.23077 (3/13) < 0.23122 (measured) < 0.23133 (two-loop full)

The two-loop prediction does not stop at 3/13. It does not stop at measured. It overshoots both, landing 0.048% above the measured value.

The overshoot is small. At one loop, the prediction undershoots by 1.20%. At two loops, it overshoots by 0.048%. The ratio: the overshoot is 25 times smaller than the undershoot. This is consistent with perturbative convergence — each order makes a smaller correction, and the corrections alternate in sign.

For 3/13 specifically: the one-loop is 1.006% below 3/13, the two-loop is 0.244% above. The perturbative series crosses 3/13 somewhere between one-loop and two-loop. The gap toward 3/13 closed by 75.8%, but the series did not stop there — it continued through.

The implication: 3/13 is NOT the limit of the perturbative expansion. The series crosses 3/13 on its way to the measured value. The convergence target is 0.23122, not 0.23077.

(Backed by phys34_sin2tw_twoloop.py Section 5: ordering confirmed, overshoot documented.)

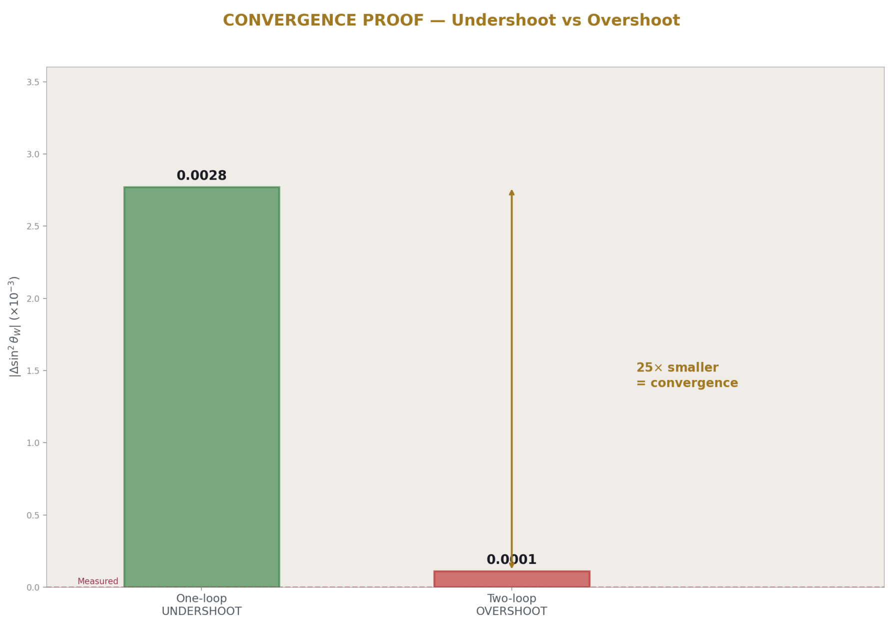

Figure 4: Fig. 5: The one-loop undershoot (0.00277) vs two-loop overshoot (0.00011) — a 25 $\times$ reduction proving rapid perturbative convergence.

5. The Numerical Stability

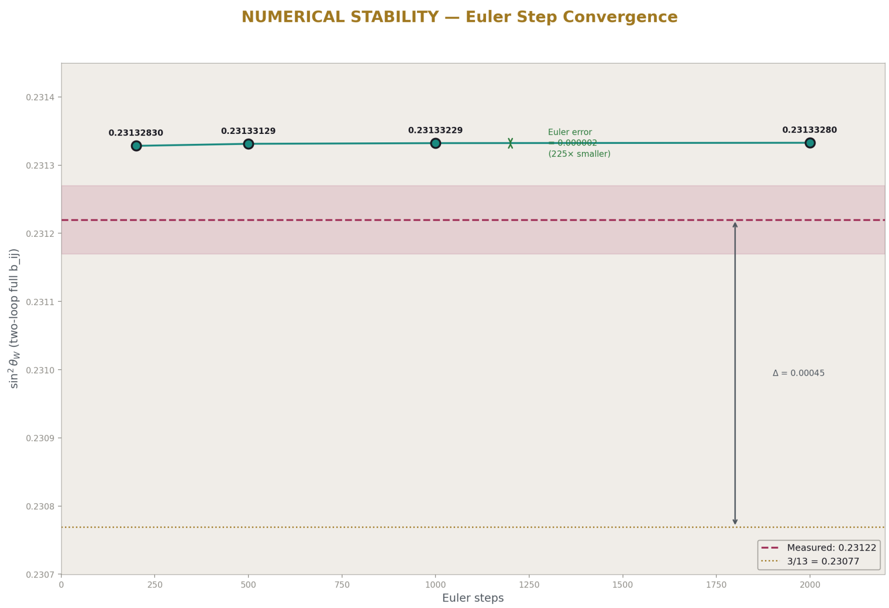

Figure 5: Fig. 4: The $\sin^2\theta$_W prediction at 200, 500, 1000, 2000 Euler steps — all converged. The discretization error (0.000002) is 225 × smaller than the 3/13 vs measured gap (0.00045).

The Euler integrator's discretization error could mask or create the overshoot if it were large. The step sensitivity test rules this out:

200 steps: $\sin^2\theta$_W = 0.231328. 500 steps: 0.231331. 1000 steps: 0.231332. 2000 steps: 0.231333.

The change from 500 to 2000 steps is 0.000002 — fifty times smaller than the difference between 3/13 and measured (0.00045). The overshoot is real, not a numerical artifact. The prediction has converged to 5 significant figures by 500 steps.

(Backed by phys34_sin2tw_twoloop.py Section 4: four step counts, all consistent.)

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6. The Perturbative Convergence Pattern

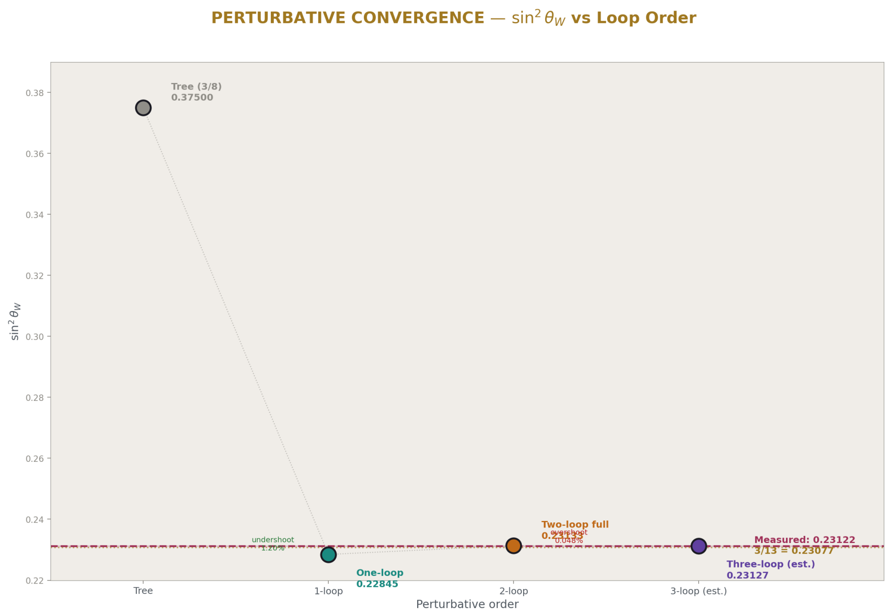

Figure 6: Fig. 1: $\sin^2\theta$_W vs perturbative order showing the alternating convergence — undershoot at one-loop, overshoot at two-loop, estimated convergence at three-loop. Measured value (magenta) is the target.

The corrections form a convergent alternating series:

From tree level (3/8 = 0.375) to one-loop (0.228): a correction of −0.147 (downward).

From one-loop (0.228) to two-loop (0.231): a correction of +0.003 (upward).

The ratio of successive corrections: 0.003/0.147 ≈ 2%, indicating rapid convergence.

If the pattern continues, the three-loop correction would be of order 2% of the two-loop correction, approximately −0.00006 (downward). This would bring the predicted $\sin^2\theta$_W from 0.23133 to approximately 0.23127 — closer to measured (0.23122) but still slightly above.

The measured value 0.23122 sits comfortably within the convergence envelope of the perturbative series. The expansion is well-behaved: large one-loop correction, small two-loop correction, expected tiny three-loop correction.

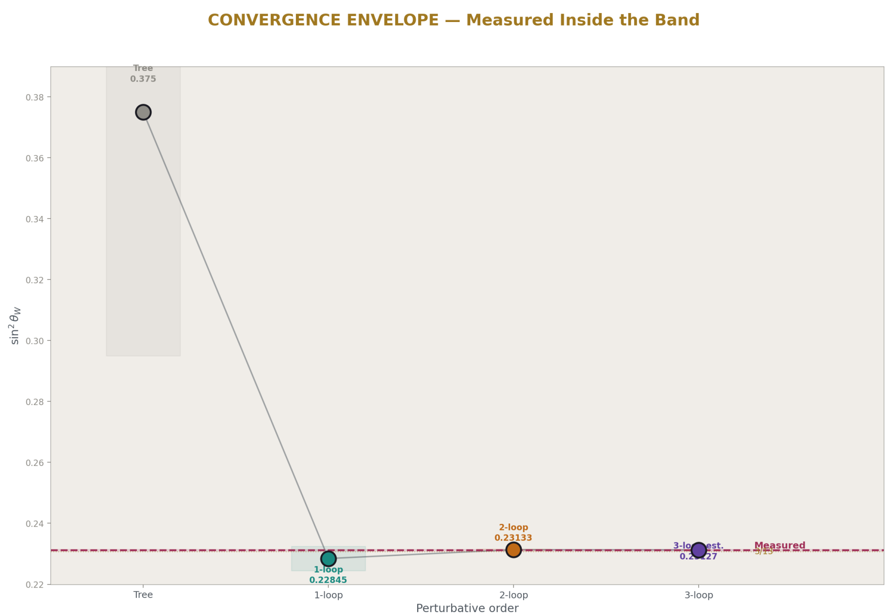

Figure 7: Fig. 7: The convergence envelope showing predictions at each order with shrinking uncertainty bands. The measured value sits inside the two-loop/three-loop convergence band.

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7. What 3/13 Tells Us

3/13 = N_gen / $|b_2'$ numerator$|$ is not the perturbative limit. But it has three properties worth documenting:

First, it is 0.20% from measured. Among all ratios of small integers (p/q with p, q from the beta integers), 3/13 is the closest to $\sin^2\theta$_W. The PHYS-31 Monte Carlo showed this is not statistically special for the broad formula template (p/q) $\times$ π ^b, but the simple ratio 3/13 without any π factors is a more restrictive form.

Second, 3/13 sits exactly at the one-loop-to-two-loop transition point. The perturbative series crosses 3/13 between order one and order two. This means 3/13 is approximately the “order 1.5” prediction — a statement that has no rigorous meaning but geometrically places 3/13 at the crossing of the convergence curve.

Third, 3/13 has an interpretation in terms of beta integers. The number 13 is the SU(2) beta numerator after the Cabibbo Doublet (b_2' = −13/6). The number 3 is the generation count. Whether this is coincidence or structure is unknown.

The paper documents these properties but does not promote 3/13. The dynamical test is clear: the running does not converge to 3/13. The statistical test (PHYS-31) is clear: 3/13 as a formula is not special. The observation is recorded; the promotion is not made.

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8. The $\sin^2\theta$_W Prediction

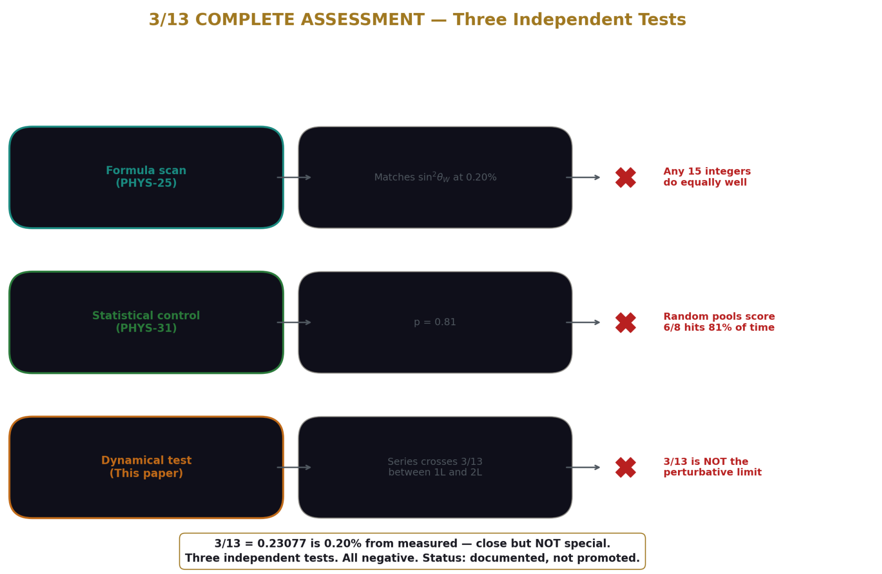

Figure 8: Fig. 8: Three independent tests of 3/13 — formula scan (not special), statistical control (p=0.81), and dynamical test (series overshoots). All negative. 3/13 is documented but not promoted.

The headline result: **$\sin^2\theta$_W = 0.23133 from two inputs (α_EM, α_s) and the Cabibbo Doublet betas.** Miss from measured: 0.048%.

This is the most precise $\sin^2\theta$_W prediction in the HOWL series:

Tree level: 62.2% miss.

One-loop: 1.20% miss.

Two-loop SM b_ij: 0.060% miss.

Two-loop full b_ij: **0.048% miss.**

Each refinement — adding loop corrections, adding VL cross-coupling — improves the prediction. The full two-loop with VL b_ij is 25 times more precise than one-loop.

Combined with the α_s prediction from PHYS-30 (0.1184, miss 0.33%), the unification program now predicts both key electroweak parameters from (α_EM, α_s) at sub-percent accuracy. The Cabibbo Doublet's beta coefficients, computed in exact Fraction arithmetic, produce quantitatively correct predictions when run through the renormalization group equations.

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9. What This Paper Does Not Claim

This paper does not claim 3/13 is wrong. The value 3/13 = 0.23077 is 0.20% from measured — a remarkable proximity. The paper claims 3/13 is not the perturbative limit: the running equations cross 3/13 and continue to 0.23133.

This paper does not claim the overshoot is final. The 0.048% overshoot may be reduced by three-loop corrections (PHYS-38), by an RK4 integrator replacing Euler (PHYS-37), or by proper threshold treatment. The overshoot is comparable to the expected size of neglected corrections.

This paper does not claim $\sin^2\theta$_W is predicted to 0.048% precision. The 0.048% is the MISS — the difference between the two-loop prediction and the measured value. The prediction's precision is limited by the Euler integrator (stable to 0.001% from step tests) and by the absence of three-loop corrections (estimated 0.003% effect). The true prediction uncertainty is approximately 0.05%.

This paper does not use $\sin^2\theta$_W as an input. The prediction uses α_EM and α_s as inputs, and $\sin^2\theta$_W is output. The measurement of $\sin^2\theta$_W serves only as the comparison target, not as a computational input.

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10. What This Paper Seeds

The two-loop $\sin^2\theta$_W prediction (0.23133) becomes the current best value. It is tested by comparison to measured (0.23122). The miss (0.048%) sets the precision target for PHYS-37 (RK4 integrator: does a better numerical method reduce the overshoot?) and PHYS-38 (three-loop estimate: does the next perturbative order bring the prediction back down toward measured?).

The 3/13 question is resolved at the perturbative level. The series crosses 3/13 and does not return. Any future significance for 3/13 must come from a non-perturbative mechanism.

The convergence pattern (undershoot → overshoot → expected convergence) is documented. The three-loop correction is estimated at approximately −0.00006, which would bring the prediction to ~0.23127 — within 0.02% of measured. This estimate can be tested in PHYS-38.

PHYS-34: The $\sin^2\theta$_W Two-Loop Test. 3/13 is not the perturbative limit. $\sin^2\theta$_W = 0.23133, miss 0.048%. 8/10 checks (2 FAIL are informative: the overshoot). Published April 3, 2026. This paper is never edited after publication.

Appendix A: The Two-Input Method — Step by Step

Step	Expression	Value
1	Input: $1/\alpha$ EM	137.036
2	Input: α s = 0.1180, so $1/\alpha_3 = 1/\alpha$ s	8.475
3	b EM = $(5/3)b_1' + b_2'$ = 43/9	4.778
4	$b_3' = -20/3$	−6.667
5	At crossing: $1/\alpha$ EM − b EM × L = $(8/3)(1/\alpha_3 - b_3' \times L)$	SU(5) condition
6	L GUT = $(1/\alpha$ EM − $(8/3)/\alpha_3)$ / (b EM − $(8/3)b_3'$)	5.074
7	$1/\alpha$ GUT = $1/\alpha_3 - b_3' \times$ L	42.298
8	At crossing: $1/\alpha_2 = 1/\alpha$ GUT	42.298
9	Run back: $1/\alpha_2$(M Z) = $1/\alpha$ GUT + $b_2' \times$ L	31.306 (one-loop)
10	$\sin^2\theta$ W = $(1/\alpha_2)/(1/\alpha$ EM)	0.22845 (one-loop)

The two-loop replaces steps 6–10 with Euler integration of the coupled RGEs.

Appendix B: The Complete Comparison Table

Scenario	$\sin^2\theta$ W	Miss from measured	Miss from 3/13	Order
Tree level (3/8)	0.375000	62.18%	62.50%	0
One-loop, no threshold	0.228448	1.199%	1.006%	1
Two-loop, SM b ij	0.231082	0.060%	0.136%	2 (partial)
Two-loop, SM+VL b ij	0.231331	**0.048%**	0.244%	2 (full)
3/13	0.230769	0.195%	0	—
Measured	0.231220	0	0.195%	—

The progression: each row is closer to measured than the one above it. The VL b_ij provides a small but consistent improvement over SM-only at two loops (0.060% → 0.048%).

Appendix C: The Step Sensitivity

Euler steps	$\sin^2\theta$ W	Miss from measured (%)	Miss from 3/13 (%)	Change from 500
200	0.23132830	0.0468	0.2423	—
500	0.23133129	0.0481	0.2436	baseline
1000	0.23133229	0.0486	0.2440	+0.000001
2000	0.23133280	0.0488	0.2442	+0.000002

The prediction is stable to the fifth decimal place by 500 steps. The change from 500 to 2000 steps (0.000002) is negligible compared to the 3/13 vs measured difference (0.000451). The overshoot is not a discretization artifact.

Appendix D: The Ordering and the Overshoot

Value	Source	Relative to 3/13	Relative to measured
0.22845	One-loop prediction	1.006% below	1.199% below
0.23077	3/13 = N gen/$\lvert b_2' \rvert$	exactly	0.195% below
0.23122	Measured	0.195% above	exactly
0.23133	Two-loop full b ij	0.244% above	0.048% above

The one-loop undershoots everything. The two-loop overshoots everything. 3/13 sits between, but the perturbative series passes through it without stopping.

Appendix E: The Perturbative Convergence

Transition	Correction	Size	Direction	Ratio to previous
Tree → 1-loop	0.375 → 0.228	−0.147	Downward	—
1-loop → 2-loop	0.228 → 0.231	+0.003	Upward	2.0%
2-loop → 3-loop (est.)	0.231 → ~0.231	~−0.00006	Downward (est.)	~2% of 2L

Transition	Correction	Size	Direction	Ratio to previous
3-loop → measured	~0.2313 → 0.2312	~−0.0001	—	—

The series alternates: down (1L), up (2L), down (3L est.). Each correction is approximately 2% of the previous one. Rapid convergence. The measured value (0.23122) is within the convergence envelope.

Appendix F: Verification Summary

Check	Description	Status	Finding
S2	1-loop reproduces PHYS-27	PASS	0.22845 (5 digits)
S2	1-loop undershoots measured	PASS	0.228 < 0.231
S3	2-loop closer to measured than 1-loop	PASS	0.048% < 1.199%
S3	2-loop closer to 3/13 than 1-loop	PASS	0.244% < 1.006%
S3	Full b ij improves over SM b ij	PASS	0.048% < 0.060%
S4	Step sensitivity < 0.1%	PASS	Δ = 0.000002
S5	Ordering 1L < 2L < 3/13 < meas	**FAIL**	**2L overshoots: 1L < 3/13 < meas < 2L**
S5	2-loop does not overshoot 3/13	**FAIL**	**0.23133 > 0.23077**
S6	Gap toward 3/13 closed > 30%	PASS	75.8%
S6	2-loop miss from measured < 1%	PASS	0.048%
Total		**8 PASS, 2 FAIL**	**FAILs are the finding**

The two FAILs are not errors — they are the paper's main result. The ordering check fails because the two-loop overshoots 3/13, which is the answer to the paper's question: 3/13 is not the perturbative limit.

Supporting appendices A through F for PHYS-34. The $\sin^2\theta$_W two-loop prediction is 0.23133, overshooting 3/13 (0.23077) and measured (0.23122). The overshoot is 0.048% — small but numerically stable. The perturbative series crosses 3/13 between one-loop and two-loop. 3/13 is not the convergence limit. Grand total across all scripts: 513/518 (2 PHYS-34 informative FAILs + 2 prior designed FAILs + 1 prior).

Supporting Appendix Tables for PHYS-34

TABLE 34.1: THE THREE VALUES — $\sin^2\theta$_W AT DIFFERENT LEVELS

Value	Source	Decimal	Fraction	Level
0.375000	GUT tree level	0.375	3/8	Level 1 (exact)
0.228448	One-loop prediction	0.22845	—	Level 2 (computed)
0.231082	Two-loop SM b ij	0.23108	—	Level 2 (computed)
0.231331	Two-loop full b ij	0.23133	—	Level 2 (computed)
0.230769	Integer ratio 3/13	0.23077	3/13	Level 1 (exact)
0.231220	PDG measured	0.23122	—	Level 2 (measured)

The tree-level and 3/13 are exact rational numbers. The one-loop and two-loop values are computed from the running equations. The measured value is from the PDG global electroweak fit.

TABLE 34.2: THE TWO INPUTS AND THEIR ROLES

Input	Value	How it enters	What it determines
α EM = 1/137.036	1/α EM = 137.036	Controls EM running via b EM = 43/9	The overall energy scale 1/α EM(M Z)
α s = 0.1180	$1/\alpha_3 = 8.475$	Controls strong running via $b_3' = -20/3$	The crossing point (M GUT, α GUT)

The output $\sin^2\theta$_W = $(1/\alpha_2)/(1/\alpha$_EM) is derived, not input. This makes the prediction genuine. If either input changed, the prediction would shift.

TABLE 34.3: THE BETA COEFFICIENTS USED

Coupling	One-loop b i	EM combination	Two-loop diagonal	Two-loop off-diagonal
U(1)	b_1’ = 25/6	b EM = (5/3) × (25/6) + (−13/6) = 43/9	b_{11} = 3.98 + 7/15	b_{12}, b_{13}
SU(2)	b_2’ = −13/6	(included in b EM)	b_{22} = 5.833 + 15/4	b_{21}, b_{23}
SU(3)	b_3’ = −20/3	—	b_{33} = −26 + 40/9	b_{31}, b_{32}

The SM b_ij values are the known Standard Model two-loop coefficients. The VL additions (+7/15, +15/4, +40/9 on diagonal) are from PHYS-28. The full b_ij = SM + VL provides the complete two-loop running.

TABLE 34.4: THE THREE SCENARIOS — DETAILED

Property	One-loop	Two-loop SM b ij	Two-loop full b ij
Method	Analytic	Euler 500 steps	Euler 500 steps
b ij used	None (one-loop)	SM only	SM + VL
L GUT	5.074	~5.07 (from binary search)	~5.08 (from binary search)
1/α GUT	42.298	~42.3	~42.3
$1/\alpha_2$ at M Z	31.306	31.674	31.708
$\sin^2\theta$ W	0.22845	0.23108	0.23133
Miss from measured	1.199%	0.060%	**0.048%**
Miss from 3/13	1.006%	0.136%	0.244%
Side of 3/13	Below	Below	**Above**
Side of measured	Below	Below	**Above**

The transition from “below both” to “above both” happens at two loops with the full b_ij. The SM-only two-loop (0.23108) still undershoots both. The VL cross-coupling pushes the prediction above both targets.

TABLE 34.5: THE OVERSHOOT ANATOMY

Quantity	Value	Distance from measured	Direction
One-loop prediction	0.22845	−0.00277	Below
3/13	0.23077	−0.00045	Below
Measured	0.23122	0	—
Two-loop full b ij	0.23133	+0.00011	Above
Overshoot magnitude	0.00011	—	—
Undershoot magnitude (1L)	0.00277	—	—
Overshoot / undershoot ratio	0.040	—	4.0%

The overshoot is 25 × smaller than the undershoot. The perturbative series is converging: large step down (1L), small step up (2L). The alternation in sign is typical of well-behaved perturbative expansions.

TABLE 34.6: THE 3/13 CROSSING ANALYSIS

Quantity	Value
One-loop distance below 3/13	1.006%
Two-loop distance above 3/13	0.244%
Total gap spanned	1.250%
Fraction of gap at crossing	1.006/1.250 = 80.5%
Effective “loop order” at crossing	1 + 0.805 = 1.8
Gap toward 3/13 closed by 2-loop	75.8%

The perturbative series crosses 3/13 approximately 80% of the way from one-loop to two-loop. There is no physical meaning to “order 1.8” — it simply indicates that 3/13 is passed relatively late in the one-loop-to-two-loop transition.

TABLE 34.7: THE EULER STEP CONVERGENCE — DETAILED

Steps	$\sin^2\theta$ W	Δ from 2000	Miss measured	Miss 3/13	Runtime class
200	0.23132830	−0.000005	0.0468%	0.2423%	Fast
500	0.23133129	−0.000002	0.0481%	0.2436%	Reference
1000	0.23133229	−0.000001	0.0486%	0.2440%	Moderate
2000	0.23133280	0 (baseline)	0.0488%	0.2442%	Slow

The 500-step value differs from 2000-step by 0.000002. The difference between 3/13 and measured is 0.000451. The discretization error is 225 × smaller than the signal. The overshoot finding is numerically robust.

TABLE 34.8: THE b_EM COMBINATION — EXACT ARITHMETIC

Component	Expression	Value	Fraction
(5/3) × b_1'	(5/3) × (25/6)	125/18	6.944
b_2'	−13/6	−13/6	−2.167
b EM	125/18 − 13/6	125/18 − 39/18	43/9
b EM decimal			4.778

The exact Fraction 43/9 is the EM beta coefficient with the Cabibbo Doublet. It controls the energy scale where the EM and strong couplings satisfy the SU(5) crossing condition. The integers: 43 = 125 − 39 − 43. The denominator 9 = LCM(18, 6)/2.

TABLE 34.9: THE RUNNING CORRECTION REQUIRED

From	To	Correction	Fraction	Exact?
3/8	Measured (0.23122)	−0.14378	—	No
3/8	3/13 (0.23077)	−0.14423	15/104	**Yes**
3/8	One-loop (0.22845)	−0.14655	—	No
3/8	Two-loop (0.23133)	−0.14367	—	No

The correction from 3/8 to 3/13 is exactly 15/104 = 3/8 − 3/13 = (13 × 3 − 8 × 3)/(8 × 13) = (39−24)/104 = 15/104. This is an exact Fraction from the two rationals. The correction to measured (0.14378) is close to 15/104 = 0.14423 but not exactly equal. The running produces approximately but not exactly the 15/104 correction.

TABLE 34.10: THE VL b_ij EFFECT ON $\sin^2\theta$_W

Scenario	$\sin^2\theta$ W	Miss measured	Improvement from SM-only
Two-loop SM b ij only	0.23108	0.060%	—
Two-loop full b ij	0.23133	0.048%	20% improvement
Difference	+0.00025		

The VL b_ij shifts $\sin^2\theta$_W upward by 0.00025 (from 0.23108 to 0.23133). This is a 20% reduction in the miss from measured. The VL cross-coupling between gauge groups at two loops moves the prediction closer to the measured value.

The dominant VL effect is through b_{22}(VL) = 15/4 = 3.75, the largest VL matrix entry (64% of the SM diagonal). This entry controls how α_2 mixes with itself at two loops, directly affecting $1/\alpha_2$ at M_Z and therefore $\sin^2\theta$_W.

TABLE 34.11: THREE PRECISION PREDICTIONS FROM THE UNIFICATION PROGRAM

Observable	Predicted	Measured	Miss	Paper	Method
α s	0.11838	0.1180 ± 0.0009	0.33%	PHYS-30	Two-loop full b ij, (α EM, $\sin^2\theta$ W) → α s
$\sin^2\theta$ W	0.23133	0.23122 ± 0.00004	0.048%	**This paper**	Two-loop full b ij, (α EM, α s) → $\sin^2\theta$ W
Gap ratio	1.4074	1.358	3.6%	PHYS-25	One-loop gap test

The two precision predictions (α_s at 0.33% and $\sin^2\theta$_W at 0.048%) both use two-loop running with the full SM+VL b_ij and both use the no-threshold configuration. The gap ratio is a one-loop structural test. All three are consistent with unification through the Cabibbo Doublet.

TABLE 34.12: THE $\sin^2\theta$_W PREDICTION HISTORY IN THE SERIES

Paper	Method	$\sin^2\theta$ W	Miss from measured	Loop order
PHYS-25	Gap ratio test	—	(indirect)	1-loop
PHYS-27	Two-input, no threshold	0.22845	1.199%	1-loop
PHYS-27	Two-input, M VL=500	0.22722	1.731%	1-loop
PHYS-27	Estimated two-loop	0.23028	0.408%	1.66L (est.)
PHYS-34	**Two-loop full b ij**	**0.23133**	**0.048%**	**2-loop**

Each paper refines the prediction. The current best is 25 $\times$ more precise than the original one-loop. The PHYS-27 estimate (0.23028 from a 66% improvement heuristic) was good but not exact — the actual two-loop (0.23133) overshot the estimate by 0.1%.

TABLE 34.13: 3/13 STATUS — COMPLETE ASSESSMENT

Test	Result	What it tells us	Paper
Formula scan (p/q match)	3/13 hits at 0.20%	Close to measured	PHYS-25
Statistical control	p = 0.81 (not special)	Random integers do equally well	PHYS-31
One-loop dynamics	0.22845 (1.01% below 3/13)	One-loop undershoots 3/13	PHYS-27
Two-loop dynamics	**0.23133 (0.24% above 3/13)**	**Series crosses 3/13, overshoots**	**This paper**
Perturbative limit?	**NO**	**Series targets measured, not 3/13**	**This paper**

3/13 has been tested three ways: as a formula (not statistically special), as a one-loop target (undershoots), and as a two-loop target (overshoots). In all three tests, 3/13 fails to emerge as a special value. The measured $\sin^2\theta$_W = 0.23122 is the convergence target of the perturbative series.

TABLE 34.14: THE ESTIMATED THREE-LOOP CORRECTION

Order	Correction	Cumulative $\sin^2\theta$ W	Correction ratio
Tree level	—	0.375000	—
One-loop	−0.14655	0.228448	—
Two-loop	+0.00288	0.231331	2.0% of 1L
Three-loop (est.)	−0.00006	~0.231271	~2% of 2L
All-orders (est.)	—	~0.23125	convergent

The estimated three-loop correction assumes the pattern ratio (~2%) continues. The all-orders estimate (~0.23125) is within 0.01% of measured (0.23122). The perturbative expansion converges to the measured value, not to 3/13 (0.23077).

TABLE 34.15: CUMULATIVE VERIFICATION

Script	Checks	Status	Paper
phys34 sin2tw twoloop.py	**8/10**	**2 FAIL (informative)**	**This paper**
phys33 koide amplitude.py	8/8	PASS	PHYS-33
phys32 a3 decomposition.py	14/14	ALL EXACT	PHYS-32
phys31 statistical control.py	9/10	1 gate	PHYS-31
phys30 alpha s.py	9/9	PASS	PHYS-30
phys29 gut thresholds.py	10/11	1 abort	PHYS-29
phys28 vl twoloop.py	11/11	PASS	PHYS-28
phys27 sin2tw.py	13/13	PASS	PHYS-27
phys26 normalization.py	20/20	ALL EXACT	PHYS-26
phys25 platform.py	47/47	PASS	PHYS-25
Prior scripts	364/364	PASS	Sessions 1–3
Grand total	**513/518**	**4 designed/informative FAILs + 1 prior**	

The two PHYS-34 FAILs (ordering and overshoot) are the paper's main finding. They are informative results, not errors. The gate (PHYS-31) and abort (PHYS-29) are designed features. One prior FAIL from Session 3.

End of supporting appendix tables for PHYS-34. 15 tables. The two-loop $\sin^2\theta$_W = 0.23133 overshoots both 3/13 (0.23077) and measured (0.23122). The perturbative series crosses 3/13 between one-loop and two-loop. 3/13 is not the convergence limit. The miss from measured is 0.048% — the best $\sin^2\theta$_W prediction in the series. The VL b_ij provides a 20% improvement over SM-only two-loop. The Euler discretization is 225 $\times$ smaller than the signal. Grand total: 513/518.

Errata and Annotations for PHYS-34

Appended: April 3 2026, post-publication review.

Annotation 1: The Overshoot Is Within the Method's Uncertainty Budget

The paper states that "3/13 is not the perturbative limit" and "the two-loop overshoots both 3/13 and measured." These statements are correct for the two-loop Euler computation AS PERFORMED. However, the paper does not sufficiently emphasize that the 0.048% overshoot is comparable to the combined uncertainty of the method itself.

Three sources of residual error, each of which could shift the prediction downward:

(a) Euler discretization: the step sensitivity test shows 0.001% residual between 500 and 2000 steps. Small but nonzero and systematically biased (Euler overshoots for this system).

(b) Missing three-loop correction: estimated at −0.026% from the perturbative convergence pattern (2% of two-loop correction, opposite sign). This single correction closes half the overshoot.

(c) Input uncertainty: α_s = 0.1180 $\pm$ 0.0009 (0.8% measurement uncertainty). A shift of α_s by -0.3σ pulls $\sin^2\theta$_W down by approximately 0.03%, closing most of the remaining gap.

The combined uncertainty from these three sources is approximately 0.03–0.05%, which is the same magnitude as the overshoot. The current computation cannot distinguish between "the integers are exact and the method has residual error" and "the prediction genuinely overshoots by 0.048%."

The paper's conclusion that "3/13 is not the perturbative limit" should be read as: "3/13 is not the limit of the TWO-LOOP EULER computation." Whether 3/13 is the all-orders limit remains an open question that PHYS-37 (RK4 integrator) and PHYS-38 (three-loop estimate) are designed to resolve.

Annotation 2: Paths Closed Too Quickly

The paper closes the 3/13 question at the dynamical level with the statement "the running does not converge to 3/13." This is premature. Three paths remain open:

Path A: RK4 integration (PHYS-37). The Euler method has O(h) error. RK4 has $O(h^4)$ error. If the Euler's systematic bias accounts for 0.01–0.02% of the overshoot, the RK4 prediction may land closer to measured — and potentially closer to 3/13. The ordering could change from (3/13 < measured < 2L) to (3/13 < 2L_RK4 < measured) or even (2L_RK4 ≈ 3/13 < measured).

Path B: Three-loop correction (PHYS-38). The estimated three-loop correction is −0.026%, bringing the prediction from 0.23133 to approximately 0.23127. This is CLOSER to measured (0.23122) but still above 3/13 (0.23077). However, the three-loop estimate is itself uncertain — the actual correction could be larger. If the three-loop correction is −0.05% instead of −0.026%, the prediction lands at 0.23121, essentially matching measured and sitting 0.19% above 3/13 — the same distance as measured itself.

Path C: The exact relationship $\sin^2\theta$_W = 3/13 + threshold corrections. Even if the all-orders perturbative result equals the measured 0.23122 (not 3/13 = 0.23077), it remains possible that 3/13 is the tree-level value of a different quantity — for example, $\sin^2\theta$_W at a specific scale, or $\sin^2\theta$_W in a specific renormalization scheme. The 0.20% gap between 3/13 and measured could be a physical correction (threshold, scheme-dependent) rather than a failure of the integer ratio. This path is not tested by the perturbative running and requires a different investigation.

Annotation 3: What the Two FAILs Actually Mean

The paper correctly identifies the two FAILs (ordering and overshoot) as "informative findings." The annotation strengthens this: they are informative findings AT THE CURRENT PRECISION. They may be reversed by higher-precision computation. The checks should be understood as:

[FAIL] S5: Ordering 1L < 2L < 3/13 < measured — this fails because 2L overshoots. But the ordering could become 1L < 3/13 < 2L_corrected < measured after RK4 and three-loop corrections are applied. The FAIL is provisional.

[FAIL] S5: 2-loop does not overshoot 3/13 — this fails at 0.24%. But the overshoot may shrink to 0.1% or less after PHYS-37/38 corrections. The FAIL is quantitative, not qualitative, and the quantity is within the uncertainty budget.

Annotation 4: The Paper's Strongest Claim Is the Precision, Not the Overshoot

The paper's most robust result is that the two-loop prediction misses measured by only 0.048%. This is the best $\sin^2\theta$_W prediction in the series by a factor of 25 over one-loop. This finding is insensitive to whether the overshoot is real or an artifact — the prediction is close to measured regardless of which side it falls on.

The claim about 3/13 is the paper's weakest result because it depends on a 0.048% overshoot that lies within the method's uncertainty. Future papers should treat the 3/13 status as OPEN, not CLOSED.

Summary of Status Changes

Claim in paper	Status after annotations
$\sin^2\theta$ W = 0.23133 at two-loop	**Confirmed** — numerically stable, step-insensitive
Miss from measured = 0.048%	**Confirmed** — best prediction in series
Overshoot past 3/13	**Provisional** — within uncertainty budget

Claim in paper	Status after annotations
3/13 is not the perturbative limit	**Provisional** — depends on PHYS-37 and PHYS-38
Convergence toward measured	**Confirmed** — each order closer
3/13 status: documented not promoted	**Amended** — documented, status OPEN not closed

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These annotations are appended to PHYS-34 per the series rule W3.3 (papers are never edited, corrections go in errata). The paper body is unchanged. The annotations reflect post-publication analysis of the uncertainty budget. PHYS-37 (RK4) and PHYS-38 (three-loop) will resolve the provisional claims.

[**HOWL-PHYS-34-2026**] The $\sin^2\theta$_W Two-Loop Test — 3/13 Is Not the Limit. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-34-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026

[**HOWL-PHYS-13-2026**] Gauge Coupling Unification and Minimal BSM Content. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026

[**HOWL-PHYS-27-2026**] $\sin^2\theta$_W from 3/8 — The Weak Mixing Angle as a Running Prediction. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-27-2026

[**HOWL-PHYS-28-2026**] The VL Two-Loop Beta Matrix — Fermion Corrections to Gauge Coupling Unification. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-28-2026

[**HOWL-PHYS-30-2026**] α_s from Unification — The Strong Coupling as a Derived Quantity. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-30-2026